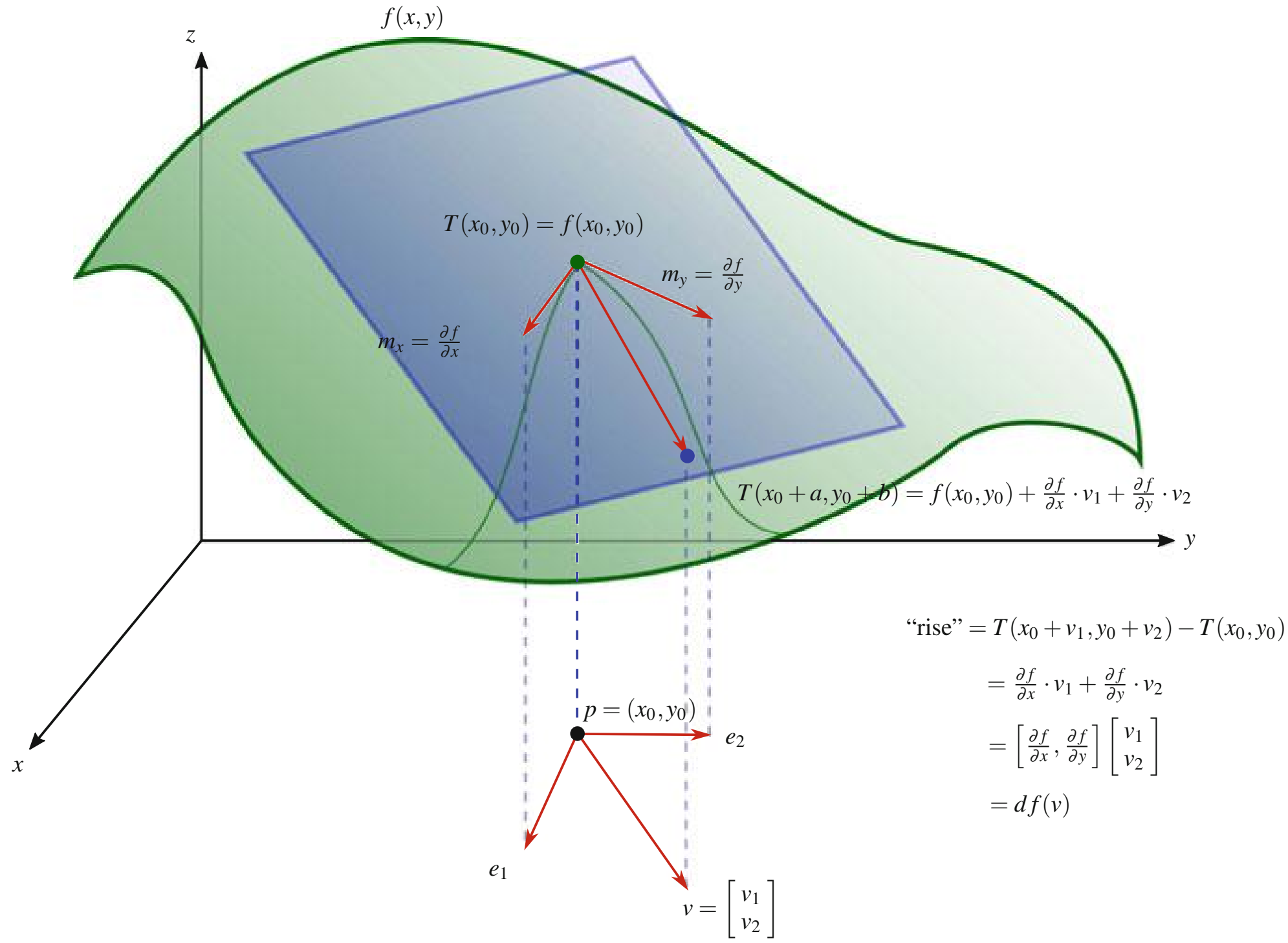
$$T(x, y) = f(x_0, y_0) + \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} \cdot (x - x_0) + \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} \cdot (y - y_0).$$



$$\begin{aligned}
\text{“rise”} &= T(x_0 + v_1, y_0 + v_2) - T(x_0, y_0) \\
&= \frac{\partial f}{\partial x} \cdot v_1 + \frac{\partial f}{\partial y} \cdot v_2 \\
&= \begin{bmatrix} \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \\
&= df(v)
\end{aligned}$$